

1. (16pts) Consider vectors  $\vec{u} = \langle 2, -1, 2 \rangle$ ,  $\vec{v} = \langle 1, 3, -1 \rangle$  and  $\hat{i} = \langle 1, 0, 0 \rangle$ . Calculate the following:

a) (5pts)  $\vec{u} \cdot \vec{v} = 2 \cdot 1 - 1 \cdot 3 - 2 \cdot 1 = -3.$

b) (6pts)  $\hat{i} \cdot (\vec{u} \times \vec{v}) = \begin{vmatrix} 1 & 0 & 0 \\ 2 & -1 & 2 \\ 1 & 3 & -1 \end{vmatrix} = 1 - 6 = -5$

OR:  $\vec{u} \times \vec{v} = \begin{vmatrix} i & j & k \\ 2 & -1 & 2 \\ 1 & 3 & -1 \end{vmatrix} = \langle 1 - 6, 2 + 2, 6 + 1 \rangle = \langle -5, 4, 7 \rangle \therefore \hat{i} \cdot (\vec{u} \times \vec{v}) = -5$

c) (5pts) Find the **scalar component** of vector  $\vec{v}$  along vector  $\vec{u}$ .

$$\text{comp}_{\vec{u}} \vec{v} = \vec{v} \cdot \hat{u} = \frac{\vec{v} \cdot \vec{u}}{|\vec{u}|} = \frac{-3}{\sqrt{4+1+4}} = \frac{-3}{3} = -1$$

2. (16pts) Consider points  $P(1, -1, -2)$ ,  $Q(-2, 0, -1)$ ,  $R(0, -2, 1)$ :

a) (8pts) Find the area of the triangle determined by the points  $P$ ,  $Q$  and  $R$ .

$$\overrightarrow{PQ} = \langle -2 - 1, 0 + 1, -1 + 2 \rangle = \langle -3, 1, 1 \rangle$$

$$\overrightarrow{PR} = \langle 0 - 1, -2 + 1, 1 + 2 \rangle = \langle -1, -1, 3 \rangle$$

$$\begin{aligned} \text{Area} &= \frac{1}{2} |\overrightarrow{PQ} \times \overrightarrow{PR}| = \frac{1}{2} \begin{vmatrix} i & j & k \\ -3 & 1 & 1 \\ -1 & -1 & 3 \end{vmatrix} \\ &= \frac{1}{2} |\langle 4, 8, 4 \rangle| = 2|\langle 1, 2, 1 \rangle| = 2\sqrt{1+4+1} = 2\sqrt{6} \end{aligned}$$

b) (8pts) Write down the equation of the plane containing these three points.

Normal is given by the cross product in part "a":  $\vec{n} = \langle 1, 2, 1 \rangle$  (or  $\langle 2, 4, 2 \rangle$ , etc.)

Equation of plane:

$$\vec{n} \cdot \vec{r} = \vec{n} \cdot \vec{r}_0$$

Use point  $Q(-2, 0, -1)$  (or any of the other two points) as position-vector  $\vec{r}_0$ :

$$x + 2y + z = -2 - 1 = -3$$

3. (8pts) Determine whether the following two lines are parallel, skew, or intersecting:

$$\text{Line 1: } x = -t, \quad y = 2t + 1, \quad z = 3. \quad \vec{v}_1 = \langle -1, 2, 0 \rangle$$

$$\text{Line 2: } x = s - 1, \quad y = -2s, \quad z = -3s. \quad \vec{v}_2 = \langle 1, -2, -3 \rangle$$

**Not parallel:** direction vectors are not proportional to each other (clear from the z-component of  $\vec{v}_{1,2}$ )

$$\text{This is also clear from the fact that } \vec{v}_1 \times \vec{v}_2 = \langle -6, -3, 0 \rangle \neq \vec{0}$$

Now, try to look for intersection:

If we equate the z coordinates, and then the x coordinates, the y coordinates won't match:

$$\begin{pmatrix} -t = s - 1 \\ 2t + 1 = -2s \\ 3 = -3s \end{pmatrix} \rightarrow \begin{pmatrix} t = 1 - s = 2 \\ 2t + 1 = -2s \\ s = -1 \end{pmatrix} \rightarrow \begin{pmatrix} t = 2 \\ 2t + 1 = 5 \neq -2s = 2 \\ s = -1 \end{pmatrix}$$

Therefore, the lines are **skew**.

4. (12pts) Consider the line  $\langle 1 + 4t, 2 + t, t \rangle$  and the plane  $x - 2y - 2z = 1$ :

a) (6pts) Determine whether the line intersects the plane.

Plug in the line equation into the plane equation to determine if they intersect:

$$x - 2y - 2z = (1 + 4t) - 2(2 + t) - 2t = -3 \neq 1$$

Thus, the line does **not** intersect the plane

b) (6pts) If they do intersect, find the intersection point. If they do not, find the distance between this line and this plane.

Given points P in the plane and Q on the line, the distance is  $d = |\overrightarrow{PQ} \cdot \hat{n}| = \frac{|\overrightarrow{PQ} \cdot \vec{n}|}{|\vec{n}|}$

Point P in the plane: set  $y = z = 0$ :  $P(1, 0, 0)$

Point Q on the line: set  $t = 0$ :  $Q(1, 2, 0) \therefore \overrightarrow{PQ} = \langle 1 - 1, 2 - 0, 0 \rangle = \langle 0, 2, 0 \rangle$

Normal to the plane:  $\vec{n} = \langle 1, -2, -2 \rangle$

Distance:

$$d = \frac{|\overrightarrow{PQ} \cdot \vec{n}|}{|\vec{n}|} = \frac{|-4|}{\sqrt{1 + 4 + 4}} = \frac{4}{3}$$

5. (12pts) Consider the planes  $3x + 2y - z = 4$  and  $2x - y - 3z = -2$ :

Plane normals are:  $\vec{n}_1 = \langle 3, 2, -1 \rangle$ ,  $\vec{n}_2 = \langle 2, -1, -3 \rangle$

- a) (8pts) Find the angle between these two planes.

$$\cos \theta = \hat{n}_1 \cdot \hat{n}_2 = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| \cdot |\vec{n}_2|} = \frac{6 - 2 + 3}{\sqrt{9 + 4 + 1} \cdot \sqrt{4 + 1 + 9}} = \frac{7}{14} = \frac{1}{2}$$

Thus, the angle is:

$$\theta = \frac{\pi}{3}$$

- b) (4pts) Find any point belonging to their intersection.

$$\text{Set } x = 0: \begin{cases} 2y - z = 4 \\ y + 3z = 2 \end{cases} \Rightarrow \begin{cases} z = 2y - 4 \\ y + 3(2y - 4) = 7y - 12 = 2 \end{cases} \Rightarrow \begin{cases} z = 0 \\ y = 2 \end{cases} \Rightarrow P(0, 2, 0)$$

$$\text{OR: set } z = 0: \begin{cases} 3x + 2y = 4 \\ 2x - y = -2 \end{cases} \Rightarrow \begin{cases} 3x + 2y = 4 \\ 4x - 2y = -4 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = 2 \end{cases} \Rightarrow P(0, 2, 0)$$

6. (12pts) Consider the following quadric surface:

$$z^2 - y^2 - 4z - 2y - x^2 = 0$$

- a) (8pts) Write this equation in standard form and identify this surface.

$$z^2 - 4z - (y^2 + 2y) - x^2 = (z - 2)^2 - 4 - [(y + 1)^2 - 1] - x^2 = 0$$

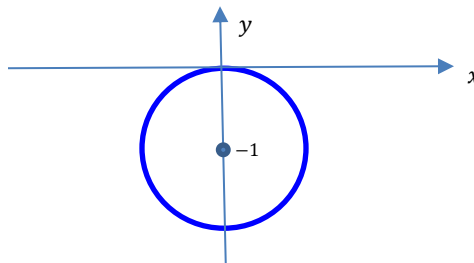
$$x^2 + (y + 1)^2 = (z - 2)^2 - 3$$

(Circular) Hyperboloid of 2 Sheets

- b) (4pts) Sketch the intersection of this surface with the plane  $z = 4$ .

$$x^2 + (y + 1)^2 = 4 - 3 = 1$$

Circle of radius 1 centered at  $(x = 0, y = -1)$ :



7. (12pts) Consider the space-curve  $\vec{r}(t) = 2t^2\hat{i} + \frac{t}{t-2}\hat{j} + (\ln t)\hat{k}$ :

a) (8pts) Find the velocity vector at  $t_0 = 1$ .

$$\frac{d\vec{r}}{dt} = 4t\hat{i} + \frac{(t-2)-t}{(t-2)^2}\hat{j} + \frac{1}{t}\hat{k} = \left\langle 4t, -\frac{2}{(t-2)^2}, \frac{1}{t} \right\rangle$$

$$\vec{v}(1) = \frac{d\vec{r}}{dt}(1) = \langle 4, -2, 1 \rangle$$

b) (4pts) Find the equation for the line tangent to this curve at  $t_0 = 1$ .

$$\vec{r}_L(s) = \vec{r}(1) + s\vec{v}(1) = \langle 2, -1, 0 \rangle + s\langle 4, -2, 1 \rangle$$

$$= \langle 2 + 4s, -1 - 2s, s \rangle$$

8. (12pts) Compute the following expressions:

a) (8pts)  $\int_0^1 [4e^{2t}\hat{i} + 3\sqrt{t}\hat{j} + \cos(\pi t)\hat{k}] dt$ :

$$\int_0^1 [4e^{2t}\hat{i} + 3\sqrt{t}\hat{j} + \cos(\pi t)\hat{k}] dt = [2e^{2t}]_0^1 \hat{i} + [2t^{3/2}]_0^1 \hat{j} + \left[ \frac{\sin(\pi t)}{\pi} \right]_0^1 \hat{k} = \langle 2(e^2 - 1), 2, 0 \rangle$$

b) (4pts)  $\frac{d}{dt}[(\vec{a} - t\vec{b}) \times (\vec{a} + t\vec{b})]$  where  $\vec{a}$  and  $\vec{b}$  are constant vectors:

$$(\vec{a} - t\vec{b}) \times (\vec{a} + t\vec{b}) = \underbrace{\vec{a} \times \vec{a}}_0 - t \underbrace{\vec{b} \times \vec{a}}_{-\vec{a} \times \vec{b}} + t \underbrace{\vec{a} \times \vec{b}}_0 - t^2 \underbrace{\vec{b} \times \vec{b}}_0 = 2t(\vec{a} \times \vec{b})$$

$$\Rightarrow \frac{d}{dt}[(\vec{a} - t\vec{b}) \times (\vec{a} + t\vec{b})] = 2(\vec{a} \times \vec{b})$$

Or:

$$\frac{d}{dt}[(\vec{a} - t\vec{b}) \times (\vec{a} + t\vec{b})] = \underbrace{\left( \frac{d}{dt}(\vec{a} - t\vec{b}) \right)}_{-\vec{b}} \times (\vec{a} + t\vec{b}) + (\vec{a} - t\vec{b}) \times \underbrace{\left( \frac{d}{dt}(\vec{a} + t\vec{b}) \right)}_{+\vec{b}}$$

$$= -\vec{b} \times \vec{a} + \vec{a} \times \vec{b} = 2(\vec{a} \times \vec{b})$$